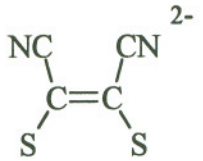
Inorganic Cumulative Exam December 6, 1997

1) The molecule NO forms two structurally distinct families of metal complexes. What are the structural differences between these series of complexes and how does the bonding differ between the series?

2) The ν(NO) is often used as a diagnostic for the nature of the M - NO interaction in metal nitrosyls. Describe the nature of this criterion and give approximate wavenumber ranges for the two types of MNO linkages.

3) The thionitrosyl ligand (NS) has also been observed. Discuss the differences between NO and NS based on the ν(CO) given below.

Compound	ν(CO)
$(\eta^5\text{-}C_5H_5)Cr(CO)_2NO$	1955, 2028 cm^{-1}
$(\eta^5\text{-}C_5H_5)Cr(CO)_2NS$	1962, 2033 cm^{-1}

4) A series of NO complexes with the formula $[Fe(mnt)_2NO]^q$ have been prepared. For the members with q = -1 and -2, which would you expect to have the higher ν(NO)? State the basis for your choice.

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